NICOLA CAMPANELLI

+39 331-303-9648 | nicola.campanelli27@gmail.com | LinkedIn | nicolacampanelli.it

Profile

Automotive Engineer with international experience in control-oriented modeling, simulation and validation of complex systems, including control optimization and energy management systems. Skilled in testing and validating models against requirements, data analysis, and technical documentation and reporting for multidisciplinary teams. Strong interest in industrial automation, digital tools and the energy transition. Reliable task finisher and disciplined, fast learner who thrives in collaborative, international settings.

Education

Politecnico di Torino 2021 - 2023
Master's Degree in Automotive Engineering *Torino, Italy*

Università degli studi di Modena e Reggio Emilia 2017 - 2020
Bachelor's Degree in Automotive Engineering *Modena, Italy*

Experience

Simulation Engineer January 2026 - current
GammaTech Engineering *Torino, Italy*

- Perform vehicle longitudinal dynamic simulations with driver-in-the-loop on a MATLAB-Simulink/GT-Suite model of a high-performance Hybrid Electric Vehicle, supporting the full process of design and development of the vehicle.
- Analyze complete powertrain behavior and perform vehicle-level energy analysis across standard homologation cycles and custom maneuvers.
- Design test maneuvers, run simulation campaigns, and post-process results to deliver them through technical reports and presentations to the project team.

Research Fellow October 2024 - July 2025
Center for Automotive Research *Columbus, OH, USA*

- Conducted a Techno-Economic Analysis of Long-Duration Battery Energy Storage Systems (BESS) to support large-scale deployment, focusing on performance, profitability, and scalability.
- Developed a MATLAB simulation model of a Distributed Energy Resources (DERs) plant, integrating an optimization control algorithm for BESS operation.
- Delivered results on the financial viability of the BESS, highlighting critical insights on its early adoption.

Research Fellow November 2023 - October 2024
Polito E3 *Torino, Italy*

- Developed a Reinforcement Learning (RL) based Energy Management System (EMS), a control strategy for a Hybrid Electric Vehicle.
- Developed a Python simulation platform for Hybrid Electric Vehicles with a focus on energy management and control strategies.
- Developed a Neural Network framework to model a physical system (vehicle HVAC), obtaining results very close to physics-based models with much lower computational time.

Skills & Tools

Software & Programming: MATLAB, Python, Git, LaTeX

Simulation & Modeling: Simulink, GT-Suite

Office Productivity Tools: Microsoft Windows, Word, PowerPoint, Excel

Languages: Italian (Native), English (C2), German (A2)

Publications

Journal paper: "Development of a Soft-Actor Critic Reinforcement Learning Algorithm for the Energy Management of a Hybrid Electric Vehicle" (DOI: 10.4271/2024-37-0011)

Journal paper: "Optimal control and techno-economic analysis of long-duration battery energy storage system applications" (DOI: 10.1016/j.est.2026.121863)

Autorizzo il trattamento dei dati personali ai sensi del Reg. UE 2016/679 (GDPR).